

# Sameer Shashikant Rajendra

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## EDUCATION

**Stevens Institute of Technology** – Hoboken, NJ

May 2026

*MS in Applied Artificial Intelligence*

**Symbiosis Skills and Professional University** – Pune, India

May 2021

*Bachelor of Technology in Mechatronics*

## TECHNICAL SKILLS

**Languages:** Python, C++, CUDA C/C++

**GPU & Parallel Computing:** CUDA, cuDNN, CUTLASS, FlashAttention-2, Triton, NCCL, NVML, NVTX, PyBind11, Nsight Systems, Nsight Compute (ncu), warp-level primitives (`__shfl_xor_sync`), shared-memory tiling, half2/bf16 vectorization

**Deep Learning Frameworks:** PyTorch (FSDP, SDPA), TensorFlow, JAX / Pallas, HuggingFace Transformers, TorchVision, timm, vLLM, DEAP

**Inference & Model Optimization:** Grouped-Query Attention (GQA), FlashAttention-2, sparse/top-k attention, KV-cache eviction, INT8 / FP8 quantization, LoRA (PEFT), kernel fusion, online softmax, autotuning (NSGA-II)

**Computer Vision:** EfficientNet, OpenCV, CuPy GPU preprocessing, Albumentations, pose estimation, multi-task real-time inference on CCTV streams, transfer learning

**Tools & Platforms:** Docker, Linux, Git, QuestDB, Jupyter, pyproject.toml / ninja parallel CUDA builds

**GPU Hardware:** NVIDIA H200 NVL (Hopper sm\_90a), H100, A100, NVLink multi-node tracing

## WORK EXPERIENCE

**Oneirix Labs Pvt. Ltd.** – Pune, India | Associate Engineer

Sept 2021 – June 2024

- Engineered a **real-time multi-task computer vision system** deployed on live Pune Metro CCTV feeds, running four concurrent inference tasks- crowd density estimation, platform intrusion and track-jump detection via **pose estimation**, property damage detection, and unattended-baggage detection — under strict per-frame latency budgets on continuous high-resolution video streams using PyTorch and TorchVision.
- Designed a deep learning model for **coronary artery diameter-reduction screening**, trading off diagnostic precision against inference overhead on resource-constrained deployment hardware for clinical use.
- Developed **low-latency C++ communication algorithms** for data exchange between controller units and scheduler subsystems; deployed containerized CV traffic-management applications via Docker, with container architecture tuned for resource efficiency and horizontal scalability.

## PROJECTS

**LLM Inference Optimization** | [GitHub](#) | [Report](#)

Jan 2026 – May 2026

- Built **three custom C++/CUDA kernels** for Llama-3.1-8B decode targeting **Hopper sm\_90a on dual NVIDIA H200 NVL**: (1) fused Grouped-Query Attention decode, (2) top-k sparse attention, and (3) fused KV-cache eviction with **INT8 / FP8 quantization**; exposed to Python via PyBind11 with a transparent PyTorch fallback.
- Reduced attention latency **16.38× at 4K** and **3.64× at 64K context** ( $115.9 \rightarrow 31.8$  ms / layer) versus FlashAttention-2 by fusing query-head grouping, KV lookup, and online softmax into a single launch with **dynamic shared-memory allocation** (bypassing the 48 KB static limit), **bank-conflict-free transposed tile\_V layout**, and half2 vectorization preserving **100% argmax generation parity** (max logit diff  $\leq 0.002$ ).
- Profiled the end-to-end pipeline with Nsight Systems / Nsight Compute, confirming a **memory-bandwidth-bound workload** ( $\sim 2.0$  FLOPs/byte) and reducing attention's GPU-time share from **51.1%  $\rightarrow$  14.4%** of total Llama-3.1-8B decode time across 32 transformer layers.
- Implemented a **PyTorch FSDP + LoRA** (rank=16,  $\alpha=32$ ) distributed training harness with NVTX-bracketed forward / backward / optimizer phases, **NCCL all-reduce overlap on H200 NVLink** multi-node, plus a **JAX / Pallas reference kernel** for cross-framework numerical verification.

## Adaptive Evolutionary GPU Optimization Framework (AEGOF) | [GitHub](#) | [Report](#) Sept – Dec 2025

- Architected an **NSGA-II multi-objective evolutionary autotuner** jointly optimizing 8 hardware / software knobs — power cap, batch size, sequence length, **bf16 vs. fp16 precision**, kernel-fusion mode, block size, tile configuration, and **FlashAttention-2 vs. CUTLASS attention backends** — for Llama-2-7B inference on **NVIDIA H100**.
- Evolved a Pareto front of **throughput vs. power** configurations across 16 individuals × 5 generations, evaluating 238+ unique configurations and reaching an **8.09× efficiency multiplier** (tokens / sec / Watt) over PyTorch defaults; results serialized to a Hall-of-Fame JSON for reproducible deployment.
- Instrumented a white-box benchmark harness measuring **tokens/sec throughput, NVML power draw, and Model FLOPs Utilization (MFU)** per trial, with robust CUDA OOM and subprocess-timeout handlers enabling unattended overnight evolutionary runs without manual intervention.

## Diabetic Retinopathy Severity Grading (EfficientNet-B4) | [GitHub](#) | [Report](#) Jan 2025 – May 2025

- Built an **EfficientNet-B4 transfer-learning pipeline** (ImageNet-pretrained backbone, fine-tuned classifier head with dropout 0.3) for 5-class diabetic retinopathy severity grading on 380×380 retinal fundus images, reaching **80.49% validation accuracy** and **0.91 F1 on the dominant class**.
- Engineered a **CuPy GPU-accelerated data loader** for high-throughput preprocessing of retinal images, and integrated sklearn-balanced class weights into nn.CrossEntropyLoss to address severe class imbalance across grades 0–4 (most patients are No-DR).

## Financial News Sentiment Classification Pipeline | [GitHub](#) | [Report](#) Jan – May 2025

- Built an end-to-end pipeline from QuestDB ingestion through GPT-Neo 125M fine-tuning to on-demand inference, processing 36,000+ financial news articles with 92% accuracy and 0.72 macro F1-score.
- Engineered a custom WeightedTrainer with class-imbalanced loss to handle severe label skew; packaged trained model and tokenizer into a streamlined inference function for real-time sentiment scoring.

## AWARDS & RECOGNITION

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### HACK MIT-WPU 2024, CCTV Video Analytics Hackathon May 2024

Maha Metro × Ministry of Education, Maharashtra. Built a real-time multi-task CCTV analytics system for Pune Metro recognized for innovation in sustainable urban transit infrastructure.